

Anaconda and Jupyter Installation Tutorial for macOS

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Introduction

This tutorial will guide you through installing Anaconda, setting up a new Conda environment, installing Jupyter, and running your first Jupyter Notebook on macOS.

1 Step 1: Downloading the Anaconda Installer

1. Visit the official Anaconda website: <https://www.anaconda.com/products/individual>.
2. On the download page, select the appropriate version for your processor:
 - If you have an **Intel** processor, download the **Intel version**.
 - If you have an **Apple M1/M2** processor, download the **Apple Silicon (M1/M2)** version.
3. Click on the download link to get the .pkg installer file for macOS.

2 Step 2: Installing Anaconda

1. Locate the downloaded .pkg file (usually in the Downloads folder).
2. Double-click the .pkg file to begin the installation.
3. Follow the on-screen prompts:
 - Click **Continue** and review the license agreement. Click **Agree** to proceed.
 - Choose the installation location (default is recommended).
4. Click **Install** and enter your system password when prompted to complete the

installation.

3 Step 3: Verifying the Installation

1. Open the **Terminal** (you can find it in Launchpad).
2. Verify the installation by typing:

```
conda --version
```

3. If this command returns the Conda version number, your installation is successful.

4 Step 4: Creating a New Conda Environment Based on the Base Environment

Conda allows you to create isolated environments for different projects. We will create a new environment that clones the base environment to ensure you have the essential packages pre-installed.

4.1 Create and Activate the Environment

1. Open the **Terminal**.
2. Create the environment by typing:

```
conda create --name myenv --clone base
```

3. Activate the environment by typing:

```
conda activate myenv
```

4.2 Explanation of the Conda Environment Setup

- `--name myenv`: Specifies the name of your new environment. In this case, it is called `myenv`, but you can choose any name.
- `--clone base`: Clones the existing base environment, ensuring your new environment starts with the same minimal packages.

4.3 Installing Jupyter

- Load your custom environment (if you have not done so previously): `conda activate myenv`.
- Install Jupyter by typing:

```
conda install jupyter
```

4.4 What is Jupyter?

Jupyter Notebook is an open-source web application that allows you to create and share documents containing live code, equations, visualizations, and narrative text. It is commonly used in data science, machine learning, and scientific computing to run Python code in an interactive, readable format.

The key features of Jupyter include:

- Executing Python code in real-time and seeing immediate outputs, such as graphs, tables, or text.
- Sharing the notebook with others so they can reproduce your analysis.
- Combining code with rich text, enabling you to explain your work and share insights interactively.

Jupyter Notebooks are widely used in education, research, and industry due to their interactive and user-friendly nature.

5 Step 5: Opening and Running Jupyter Notebook

5.1 Starting Jupyter Notebook

1. In the **Terminal**, make sure the environment is active: `conda activate myenv`.
2. Start Jupyter Notebook by typing:

```
jupyter notebook
```

5.2 Creating a New Notebook

1. In the Jupyter interface, click on **New** (upper-right corner).
2. Choose **Python 3** under the “Notebook” section.

5.3 Running Python Code in Jupyter Notebook

1. In the new notebook, you will see an empty code cell that looks like this:
`In [ ]:.`
2. Click inside the cell and type `print('Hello, World!')` and press `Shift + Enter` to run it.

5.4 Saving Your Work

To save your notebook, click **File** → **Save Notebook** or press `Command + S`.

6 Step 6: Closing the Jupyter Notebook

1. Save your work if you have not already.
2. To close the notebook, click **File** → **Shutdown**.
3. Stop the Jupyter server by closing the Terminal or pressing `Control + C`.